

DIMAS MAULANA FASYA

M.S. Candidate (Architecture) – Building Energy Performance & Sustainable Tropical Design
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RESEARCH INTEREST

Building energy performance simulation | Passive and hybrid envelope systems | Natural and buoyancy-driven ventilation | Double-skin facades in tropical climates | thermal-hygric decoupling | Computational building science

EDUCATION

- 2024 – present (Exp. Aug 2026) **Master of Science (M.S)**
Department of Architecture, College of Engineering, Pusan National University, Busan, South Korea
Thesis: Energy Performance Optimization of Integrated Passive Envelope for Mid-Rise Office Buildings in Tropical Climate
Advisor: Prof. Seonghwan Yoon
GPA: 4.31 / 4.50
- 2013 - 2017 **Sarjana Arsitektur (S.Ars) (Equivalent to Bachelor of Architecture)**
Department of Architecture, Faculty of Science and Technology, Maulana Malik Ibrahim State Islamic University, Malang, East Java, Indonesia
Final Project Entitled: Design of Riau Education Center for Forest Fire Disaster Mitigation with Eco-Sustainable Architecture Approach
GPA: 3.73 / 4.00 (Cum Laude)
Achievement: Best Graduate of the Department of Architecture in 2017

RESEARCH EXPERIENCE

- September 2024 - present **Building Environment Science and Technology (Lab of the BEST), Department of Architecture, College of Engineering, Pusan National University – Full Time Researcher**
IRIS: 13278025
KRI: 164.125.222.174
- October 2016 – January 2017 **Optimizing Roof Material as The Refrigeration and Acoustic Factor on Residential Home Building in Tropical Area - Building Science Laboratory, Department of Architecture, Maulana Malik Ibrahim State Islamic University – Research Assistant**
 - Research focuses on the influence of building roof covering materials that will be used on the thermal and acoustic conditions of residential homes in tropical climates.
- October 2015 – January 2016 **Optimizing Masterplan Development of Campus 3 of Maulana Malik Ibrahim State Islamic University, Malang Low Impact Development Based - Building Science Laboratory, Department of Architecture, Maulana Malik Ibrahim State Islamic University – Research Assistant**
 - Research focuses on the influence of building materials (specifically building facades) that will be used on building thermal and acoustic conditions in the Development of Campus 3 Maulana Malik Ibrahim State Islamic University

WORK EXPERIENCE

- June 2022 – present **Studio Humanis, Pekanbaru, Riau – Deputy Head of Studio**
Responsible for architectural design team leader, particularly in the exterior. I am in charge of meeting and discussing with clients and visualizing or rendering designs. I ensure that all our plans are presented

in detail and according to the agreed technical specifications. Since our company has been established for approximately nine months, I have offered exterior design services for sixteen (16) and twelve (12) interior projects.

April 2022 – August 2023 **PT. Alva Kreatif Industri, Pekanbaru, Riau – Senior Architect**

Responsible as a senior architect with project coverage of both interior and exterior. During my work, I handled twenty (20) interior and five (5) exterior projects.

November 2021 – March 2022 **PT. Galepala Propertindo, Pekanbaru, Riau – Senior Architect**

As an architect, I am responsible for interior and exterior projects consisting of residential houses, restaurants, and hospitals. During the approximately five months I worked, I completed around five (5) exterior and six (6) interior projects.

April 2020 – January 2022 **Monthly Bulletin of Indonesian Institute of Architects of Riau Province (BULIR), Pekanbaru, Riau – Layout Editor & Distributor**

Responsible and focused on the design and layout of the newsletter section and advertisements

July 2020 – November 2021 **FCS Indonesia, Pekanbaru, Riau – Senior Architect**

Responsible and focused on residential/housing, interior, office building, commercial building, landscape, and masterplan projects. I have completed nine (9) exterior and six (6) interior projects as a senior architect.

February 2018 – June 2020 **Kyuu Architecture Managed by PT. Ardeko Indonesia, Pekanbaru, Riau - Architect**

Responsible and focused on residential/housing, interior, office building, commercial building, landscape, and masterplan projects. During approximately two years of my employment as an architect, I completed seventeen (17) exteriors and thirteen (13) interior projects.

September 2017 – February 2018 **Angkasa Architects, Pekanbaru, Riau – Junior Architect**

Responsible and focused on residential, interior, and commercial building projects. As a junior architect for approximately seven months, I have completed four (4) exterior and four (4) interior projects.

AWARDS

2018 **Top 30th Best Essay “Essay Pemuda Competition” Organized by STUDEC International – National Scale**

2016 **6th Place Best Proposal of Student Competitive Research (PKM) Organized by Faculty of Science and Technology, Maulana Malik Ibrahim State Islamic University, Malang, East Java – Intra-University Scale**

2015 **3rd Place in “City Design Competition” – General Category – Organized by The Governments of Malang City, East Java – Regional Scale – Team Name: “Splendid”**

FUNDING

2023 - 2026 **Global Korean Scholarship (GKS) – Funded by National Institute for International Education (NIIED) – Government of Republic of Korea**

2024 - 2026 **BK21 FOUR – Funded by National Research Foundation of Korea (Research and Education Program for Living SOC Innovative Design Using Empty Spaces in Declining Cities)**

CONFERENCES

2025 **The 15th International Conference on Green Technology 2025 – Faculty of Science and Technology, Maulana Malik Ibrahim State Islamic University, Malang, Indonesia**

Presentation/Paper Title: Optimization of Hybrid Cooling System for Tropical Mid-Rise Office Building in Jakarta, Indonesia

2025 **International Symposium of Future Earth Collaboration of Pusan National University (PNU) G-LAMP and Laboratory of Architectural Science and Technology – Sepuluh Nopember Institute of Technology, Surabaya, Indonesia**

Presentation / Paper Title: A study on the Impact of a Hybrid Cooling System on Indoor Air Quality in a Tropical Mid-Rise Office Building in Jakarta, Indonesia

- 2025 **The International Conference by Indonesian Students in Korea (ICONIK) held in Kangwon National University, Chuncheon, South Korea**
Presentation / Paper Title: Optimizing Hybrid Cooling Strategies for Carbon Neutrality in Tropical Mid-Rise Office in Jakarta, Indonesia: Integrating Windcatcher and Louvers in Double Skin Façade System

PUBLICATION

- 2025 **Paper / Journal Title: Assessing the Potential of Hybrid Cooling Systems for Energy Efficiency in Tropical Mid-Rise Office Building under Jakarta's Climate Condition** – IOP Conference Series: Earth and Environmental Science (Scopus Index)
- 2026 **Paper / Journal Title: Thermodynamic Limits of Integrated Passive Envelopes in High-Entalphy Tropical Climates: Decoupling Sensible Cooling Efficiency From Latent Load Constraints** – Energy & Environment (Sage Journals) (Q1 SSCI) – Status: Awaiting Reviewer Assignment (Under Review)
- 2026 **Paper / Journal Title: Aerodynamic and Hygric Limits of Integrated Passive Envelope in Tropical Climate: An Airflow Network Diagnostic Framework for Mid-Rise Office Buildings** – Energy Reports (Elsevier) (Q1 SSCI) – Status: Under Review

RESEARCH PROJECT

- 2026 **Solar Microclimate Canopy Development (Team “Linkway”)** Funded by PNU BK21 Innovation Course
Description: Developed an AI-driven, aero-kinematic building-integrated photovoltaic (BIPV_ canopy system – integrated Edge AI (ESP32) and dual-function solar sensors to automate ventury-eefect passive cooling and stack effect thermal purging for urban pedestrian comfort

ARCHITECTURAL COMPETITION EXPERIENCE

- 2021 **Participants in Rendering Challenge 2021 organized by Andy Rahman Architects, DEMIX, and SketchUnesia – National Scale**
- 2019 **Participants in the Nusantara Architectural Design Competition Tourism Information Center organized by PT. Propan Raya – National Scale**
- The Design Object is the Tourism Information Center of Lake Toba, North Sumatera
- 2019 **Participants in the Mitra10 IDEA Architecture Design Competition themed “Contemporary Small House” organized by Mitra10 and IDEA – National Scale**
- The Design Object is a house design on a 72 m² design site with a maximum building area of 60 m²
- 2019 **Participants in the Onduline Green Roof Award #4 with the theme “Tropical Green Roof System” organized by Onduline – National Scale**
- The Design Objects are commercial buildings consisting of a capsule hotel and eatery
- 2018 **Participants in the Mitra10 Architecture Design Competition themed “Tropical Contemporary House” organized by Mitra10 – National Scale**
- 2018 **Participants in the West Bandung Regency Gate Design Competition organized by the Government of West Bandung Regency in collaboration with the Indonesian Institute of Architects of West Java Province – National Scale**
- 2017 **Participants in the Architectural Design Idea of Public Park Toilet Building Competition organized by PT. American Standard Indonesia – National Scale**

PROFESSIONAL MEMBERSHIP

The Indonesian Institute of Architects (Ikatan Arsitek Indonesia) (Ordinary Member)
Registered Number: 19546 192 1200

ORGANIZATION

ACTIVITIES

- 2023 - 2025 **The Indonesian Institute of Architects of Riau Province - Board Member – Division of Mediation and Advocacy**
- 2019 - 2022 **The Indonesian Institute of Architects of Riau Province - Board Member – Division of Organizational, Institutional, and International Relations**
As a division member, I am responsible for publishing organizational information to social media and public relations.

TRAINING, WORKSHOP, AND FORUM

- January 25th, 2023 **Architect Professional Development 2: Architectural and Environmental Planning & Multidisciplinary Project Management** organized by the Indonesian Institute of Architects of South Sumatera Province
- April 18th, 2022 **Building Structure Design Workshop for a Contoured Land** organized by the Department of Architecture, Maulana Malik Ibrahim State Islamic University
Keynote Speaker: Prof. Andrew Charleson (Victoria University of Wellington, New Zealand)
- March 11th, 2022 **Building Structure Design Workshop for Disaster Resilience** organized by the Department of Architecture, Maulana Malik Ibrahim State Islamic University
Keynote Speaker: Prof. Andrew Charleson (Victoria University of Wellington, New Zealand)
- July 16th, 2020 **Upgrading Strata 2 – Urban Planning** organized by the Indonesian Institute of Architects of Riau Province
- July 15th, 2020 **Upgrading Strata 1 – Land Use** organized by the Indonesian Institute of Architects of Riau Province
- January 26th, 2020 **Upgrading the Ethics Code and Norms of Architect Profession Attitude** organized by the Indonesian Institute of Architects of Riau Province

CERTIFICATION

- February 3rd, 2026 **Certificate of Competence – Indonesian Professional Certification Authority (Construction Services Development Board)**
Competency: Assistant Architect (Expert) Lv. 7
Certificate Number: 85691 2161.01 7 00000048 2026
No. Reg.: F 2377 00048 2026 0568535 AR 01

PROFESSIONAL/ TECHNICAL SKILLS

- Technical Skills: AutoCAD, DesignBuilder + EnergyPlus, OpenStudio, Rhinoceros + Grasshopper
- Design Skills: SketchUp, V-ray Rendering for SketchUp, Enscape, Adobe Photoshop, Adobe InDesign, Adobe Illustrator, and CorelDraw
- Video Editing: Adobe Premiere
- Document Preparation: Microsoft Office, OriginPro

LANGUAGES

- Bahasa Indonesia: **Native**
- English: **Upper-Intermediate - (TOEFL ITP: 513) (2022)**
Listening Comprehension : 50
Structure & Written Expression : 51
Reading Comprehension : 53
- Upper-Intermediate – (TOEIC: 710) (2026)**
Listening Comprehension : 400
Reading Comprehension : 310

Korea: **TOPIK 2: Level 3 (2024)**

Listening : 64/100

Writing : 27/100

Reading : 36/100

REFERENCES

- **Prof. Seonghwan Yoon (Advisor) – Master Degree**

Department of Architecture, College of Engineering, Pusan National University

Professor

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- **Aldrin Yusuf Firmansyah, M.T. (Supervisor) – Undergraduate Degree**

Department of Architecture, Maulana Malik Ibrahim State Islamic University, Malang, East Java –

Lecturer

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Email: aldrinfirmansyah@yahoo.com

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- **Ernaning Setyowati, M.T. (Laboratory Supervisor) – Undergraduate Degree**

Department of Architecture, Maulana Malik Ibrahim State Islamic University, Malang, East Java - Head
of Building Science Laboratory / Lecturer

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